



Kahoot! Unsaved changes

Quiz 14 - Directional Drilling

Directional drilling consists of well planning, anti-collision, BHA, torque and drag, and M/LWD by Sahat Martua from Sperry DD Technical Advisor

Questions (20)

1 - Quiz

Which of the following best describes a horizontal well based on inclination?

▲ Inclination less than 3°	◆ Inclination greater than 86° ✓
● Inclination equal to 45°	■ Inclination between 30° and 60°

2 - Quiz

What directional measurement indicates the direction of a deviated well?

▲ Depth	◆ Inclination
● Azimuth ✓	■ Radius of Curvature

3 - True or false

Check the question on the image, true or false?

In well planning optimization, when the wellbore is intentionally deviated at a shallow depth, the practice is called a "nudge"

◆ True ✓	▲ False
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4 - Quiz

What are ECD and SPP associated with in well profile design?

▲ Casing selection criteria	◆ Formation damage
● Hydraulic requirements and limitations ✓	■ Bit wear estimation

5 - Quiz

Why should curvature (build, drop, and turn rates) be minimized in well design?

▲ To reduce drilling speed	◆ To simplify the trajectory and reduce mechanical risk ✓
● To increase torque	■ To increase directional control

6 - Quiz

What is "slip velocity" in the context of drilling operations?

▲ The speed of pipe rotation	◆ The rate of mud loss into the formation
● The downward velocity of cuttings through the drilling fluid ✓	■ The velocity of casing during running operations

7 - Quiz

What should you do if you suspect a collision based on rough drilling and want to investigate further?

▲ Increase pump rate to flush the hole	◆ Circulate bottoms-up and check for metal or cement cuttings ✓
● Shut down the well immediately	■ Proceed with drilling at a faster rate to clear the blockage

8 - Quiz

Which of the following may indicate magnetic interference due to a nearby wellbore or metal obstruction?

▲ High annular pressure	◆ MWD survey inconsistencies ✓
● Torque fluctuations	■ Mud weight instability

9 - Quiz

The ellipsoid of uncertainty (3D) includes errors in which three primary directional survey components?

▲ Mud weight, torque, and RPM	◆ Hook load, pressure, and bit rotation
● Inclination, azimuth, and measured depth ✓	■ Dogleg severity, WOB, and circulation rate

10 - Quiz

Which type of BHA relies on rotating the entire drillstring for directional control?

▲ Steerable BHA	◆ Rotary BHA ✓
● Combination BHA	■ Bent housing BHA

11 - Quiz

What type of friction occurs when two surfaces are not moving relative to each other?

▲ Kinetic friction	◆ Static friction ✓
● Torsional friction	■ Hydrodynamic friction

12 - Quiz

Torsional friction (torque) acts in which direction relative to the drillstring?

▲ Opposite to the direction of pipe rotation ✓	◆ Along the axis of the pipe ✓
● In the same direction as mud flow	■ In a random direction

13 - True or false

A lower friction factor is typically observed in cased holes with oil-based mud.

◆ True ✓	▲ False
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14 - Quiz

What are the three stages of buckling in a drillstring?

▲ Linear, angular, and circular	◆ Smooth, rough, and locked
● Static, kinetic, and dynamic	■ Sinusoidal, helical, and lock-up ✓

15 - True or false

Increasing the stiffness of the drillstring helps in reducing the risk of buckling.

◆ True ✓ ▲ False

16 - Quiz

Which friction factor range is typical for an open hole with water-based mud?

▲ 0.15–0.20 ◆ 0.10–0.12
● 0.30–0.35 ✓ ■ 0.22–0.25

17 - Quiz

Which of the following reduces the likelihood of buckling in the drillstring?

▲ Decreasing stiffness (EI) ◆ Adding lubricants to reduce wellbore friction ✓
● Increasing radial clearance ■ Using dry drilling techniques

18 - Quiz

Which of the following is NOT a typical application of torque and drag calculations?

▲ Casing design ◆ Well profile comparison
● Mud formulation selection ✓ ■ On-site planning for sidetracks

19 - Quiz

One key benefit of MWD is:

▲ Allows automatic drilling ◆ Delivers real-time directional and toolface data ✓
● Provides delayed formation evaluation ■ Removes the need for casing design

20 - True or false

Torque and drag models are used only during well planning and not during drilling

◆ True ▲ False ✓

Details

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